

What are the pricing plans available for your software?

```
(venv) PS D:\Downloads\customer_support_system> python demo.py
[Memory] SQLite database ready at: D:\Downloads\customer_support_system\customer_support_system\memory.db
[System] Starting AI Customer Support System...
[Graph] LangGraph workflow compiled successfully.
[System] System ready.

-----

ABC TECHNOLOGIES - AI CUSTOMER SUPPORT SYSTEM
Built with LangGraph

-----

Please enter your details to begin.

Your name      : Harshita
Your customer ID: 1

Welcome, Harshita! (ID: 1)
Type your support query below. Type 'quit' to exit.

-----

Your query : What are the pricing plans available for your software

[System] Processing your query...

[Intent] Classifying query: 'What are the pricing plans available for your software'
[Memory] Retrieved 0 history entries for '1'.
[Intent] Classified as: Sales
[Router] Intent='Sales' + routing to 'sales_agent'
[Sales Agent] Processing query...
[RAG] Loaded 4 knowledge-base documents.
[RAG] Created 30 text chunks.
[RAG] Loading HuggingFace embedding model (all-MiniLM-L6-v2)...
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster down
loads.
Loading weights: 100% | 103/103 [00:00<00:00, 3341.58it/s]
[RAG] FAISS vector store built successfully.
[RAG] Retrieved 4 context chunks for query.

[Supervisor] Reviewing draft response...

PROBLEMS DEBUG CONSOLE PORTS TERMINAL
python + ~ [] [] ... | {} X

[Supervisor] Reviewing draft response...
[Supervisor] Draft passes all quality checks ✓
[Supervisor] Notes: Draft approved without modifications.
[Memory] Saved interaction for customer '1'.
[Supervisor] Interaction saved to memory for customer '1'.

=====

Customer   : Harshita (1)
Query      : What are the pricing plans available for your software
Intent     : Sales
Department : Sales
Supervisor : Draft approved without modifications.

-----
RESPONSE:
-----

Hello Harshita,

Thank you for contacting ABC Technologies Sales.

I understand your query: "What are the pricing plans available for your software"

Based on our knowledge base, here is the information relevant to your request:

[Source: pricing_guide]
ABC TECHNOLOGIES - PRICING GUIDE
=====

1. SUBSCRIPTION PLANS
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STARTER PLAN - $29/month (or $290/year - save 17%)
- Up to 5 users
- 10 GB cloud storage
- Basic reporting
- Email support
- Core business management features
- Mobile app access

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[Source: pricing_guide]
PROFESSIONAL PLAN - $79/month (or $790/year - save 17%)
- Up to 25 users
```

```
[Source: pricing_guide]
PROFESSIONAL PLAN - $79/month (or $790/year - save 17%)
- Up to 25 users
- 50 GB cloud storage
- Advanced reporting & analytics
- Priority email & chat support
- All Starter features
- API access
- Custom integrations
- Automated workflows
```

```
[Source: pricing_guide]
ENTERPRISE PLAN - $199/month (or $1,990/year - save 17%)
- Unlimited users
- 500 GB cloud storage
- Custom reporting
- 24/7 dedicated support
- All Professional features
- Custom SLA
- On-premise deployment option
- Dedicated account manager
- SSO / SAML integration
- Advanced security features
```

```
[Source: pricing_guide]
2. ADD-ONS
-----
- Extra Storage: $10/month per 10 GB
- Additional Users (Starter): $5/user/month
- Advanced Analytics Module: $30/month
- Custom Branding: $20/month
- Priority Support Upgrade: $25/month
```

```
3. FREE TRIAL
-----
- 14-day free tri
```

Please do not hesitate to reach out if you need further assistance.

Please do not hesitate to reach out if you need further assistance.

Best regards,
Alex | Sales Specialist
ABC Technologies Sales

=====

Do you have another query? (yes/no) : no

=====

Thank you for contacting ABC Technologies, Harshita!
Total queries handled this session: 1
Your conversation history is saved under ID: 1

=====

(venv) PS D:\Downloads\customer_support_system\customer_support_system> |

QUERY -2 :

I forgot my account password.

```
PROBLEMS DEBUG CONSOLE PORTS TERMINAL
```

powershell + - [] [] ... | [] [] X

```
(venv) PS D:\Downloads\customer_support_system\customer_support_system> python demo.py
[Memory] SQLite database ready at: D:\Downloads\customer_support_system\customer_support_system\memory.db

[System] Starting AI Customer Support System...
[Graph] LangGraph workflow compiled successfully.
[System] System ready.

=====

ABC TECHNOLOGIES – AI CUSTOMER SUPPORT SYSTEM
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=====

Please enter your details to begin.

Your name      : Harshita
Your customer ID: 1

Welcome, Harshital (ID: 1)
Type your support query below. Type 'quit' to exit.

-----

Your query : I forgot my password

[System] Processing your query...

[Intent] Classifying query: 'I forgot my password'
[Memory] Retrieved 6 history entries for '1'.
[Intent] Classified as: Account
[Router] Intent='Account' → routing to 'account_agent'
[Account Agent] Processing query...
[RAG] Loaded 'company_policy' as 1 whole-file document.
[RAG] Loaded 'pricing_guide' as 1 whole-file document.
[RAG] Loaded 'technical_manual' as 1 whole-file document.
[RAG] Loaded 'faq' as 28 Q&A chunks.
[RAG] Loaded 31 total knowledge-base documents/chunks.
[RAG] 28 pre-chunked (FAQ) docs + 22 character-split chunks = 50 total chunks.
```


My application crashes whenever I upload a file.

```
(venv) PS D:\Downloads\customer_support_system\customer_support_system> python demo.py
[Memory] SQLite database ready at: D:\Downloads\customer_support_system\customer_support_system\memory.db

[System] Starting AI Customer Support System...
[Graph] LangGraph workflow compiled successfully.
[System] System ready.

=====

ABC TECHNOLOGIES — AI CUSTOMER SUPPORT SYSTEM
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=====

Please enter your details to begin.

Your name      : Raju
Your customer ID: 2

Welcome, Raju! (ID: 2)
Type your support query below. Type 'quit' to exit.

-----

Your query : My application crashes whenever I upload a file.

[System] Processing your query...

[Intent] Classifying query: 'My application crashes whenever I upload a file.'
[Memory] Retrieved 0 history entries for '2'.
[Intent] Classified as: Technical
[Router] Intent='Technical' → routing to 'technical_agent'
[Technical Agent] Processing query...
[RAG] Loaded 'company_policy' as 1 whole-file document.
[RAG] Loaded 'pricing_guide' as 1 whole-file document.
[RAG] Loaded 'technical_manual' as 1 whole-file document.
[RAG] Loaded 'faq' as 28 Q&A chunks.
[RAG] Loaded 31 total knowledge-base documents/chunks.
[RAG] 28 pre-chunked (FAQ) docs + 22 character-split chunks = 50 total chunks.
[RAG] Loading HuggingFace embedding model (all-MiniLM-L6-v2)...
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster down loads.
Loading weights: 100%|██████████████████████████████████████████████████████████████████████████████| 103/103 [00:00<00:00, 7389.90it/s]
[RAG] FAISS vector store built successfully.
[RAG] Retrieved 1/4 chunks above threshold (0.5) for query: 'My application crashes whenever I upload a file.'
    [kept] score=0.5095 source=faq
    [dropped] score=0.6924 source=technical_manual
```



```
[dropped] score=0.8772 source=technical_manual
[dropped] score=1.1376 source=faq
[RAG] Retrieved 1/2 chunks above threshold (0.5) for query: 'file upload troubleshooting application error'
[kept] score=0.6897 source=technical_manual
[dropped] score=0.8142 source=technical_manual
```

```
[Supervisor] Reviewing draft response...
[Supervisor] Draft passes all quality checks ✓
[Supervisor] Notes: Draft approved without modifications.
[Memory] Saved interaction for customer '2'.
[Supervisor] Interaction saved to memory for customer '2'.
```

```
=====
Customer   : Raju (2)
Query      : My application crashes whenever I upload a file.
Intent     : Technical
Department : Technical Support
Supervisor : Draft approved without modifications.
```

```
-----
RESPONSE:
-----
```

Hello Raju,

Thank you for contacting ABC Technologies Technical Support.

I understand your query: "My application crashes whenever I upload a file."

Based on our knowledge base, here is the information relevant to your request:

[Source: faq]

Q: Why is my application crashing?

A: Common causes include browser cache issues, unsupported file formats, or files exceeding size limits. Try clearing your cache or using a different browser.

--- Additional Troubleshooting ---

[Source: technical_manual]

4. FILE UPLOAD TROUBLESHOOTING

Common causes of file upload crashes:

- File size exceeds the plan limit (Starter: 25MB, Pro: 100MB, Enterprise: 500MB)
- Unsupported file format (supported: PDF, DOCX, XLSX, PNG, JPG, CSV, ZIP)
- Slow internet connection causing timeout
- Browser cache/cookies issue

Please do not hesitate to reach out if you need further assistance.

Best regards,

Best regards,
Sam | Technical Support Engineer
ABC Technologies Technical Support

```
=====
Do you have another query? (yes/no) : no
=====
```

Thank you for contacting ABC Technologies, Raju!
Total queries handled this session: 1
Your conversation history is saved under ID: 2

```
=====
(venv) PS D:\Downloads\customer_support_system\customer_support_system> |
```

QUERY -4:

I need a refund for my annual subscription.

```
(venv) PS D:\Downloads\customer_support_system\customer_support_system> python demo.py
[Memory] SQLite database ready at: D:\Downloads\customer_support_system\customer_support_system\memory.db

[System] Starting AI Customer Support System...
[Graph] LangGraph workflow compiled successfully.
[System] System ready.

=====

ABC TECHNOLOGIES – AI CUSTOMER SUPPORT SYSTEM
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=====

Please enter your details to begin.

Your name      : Pranay
Your customer ID: 3

Welcome, Pranay! (ID: 3)
Type your support query below. Type 'quit' to exit.

-----
Your query : I need a refund for my annual subscription.

[System] Processing your query...

[Intent] Classifying query: 'I need a refund for my annual subscription.'
[Memory] Retrieved 0 history entries for '3'.
[Intent] Classified as: Billing
[Router] Intent='Billing' → routing to 'billing_agent'
[Billing Agent] Processing query...
[RAG] Loaded 'company_policy' as 1 whole-file document.
[RAG] Loaded 'pricing_guide' as 1 whole-file document.
[RAG] Loaded 'technical_manual' as 1 whole-file document.
[RAG] Loaded 'faq' as 28 Q&A chunks.
[RAG] Loaded 31 total knowledge-base documents/chunks.
[RAG] 28 pre-chunked (FAQ) docs + 22 character-split chunks = 50 total chunks.
[RAG] Loading HuggingFace embedding model (all-MiniLM-L6-v2)...
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster down
loads.
Loading weights: 100% | 103/103 [00:00<00:00, 7761.93it/s]
[RAG] FAISS vector store built successfully.
[RAG] Retrieved 1/4 chunks above threshold (0.5) for query: 'I need a refund for my annual subscription.'
      [kent] score=0.9599 source=faq
      [dropped] score=1.1980 source=faq
[Billing Agent] ⚠ HIGH-RISK request detected — flagging for human approval.
[Router] High-risk request detected → human approval required.

=====
🔴 HUMAN-IN-THE-LOOP APPROVAL REQUIRED
=====
Customer ID : 3
Customer Name: Pranay
Intent      : Billing
Department  : Billing

Customer Query:
  I need a refund for my annual subscription.

Draft Response (for supervisor review):
-----
Hello Pranay,

Thank you for contacting ABC Technologies Billing Support.

I have received your request regarding: "I need a refund for my annual subscription."

This request requires review and approval from our billing supervisor before it can be processed. Your case has been escalated and
you will receive a response within 24 business hours.

Reference information from our policy:

[Source: faq]
Q: How do I request a refund?
A: Submit a refund request through the support portal. Refunds within 30 days are processed automatically; others are reviewed manu
ally within 5 business days.

Best regards,
Jordan | Billing Support Specialist
ABC Technologies Billing
-----

Supervisor Decision – type 'approve' or 'reject': approve

[Human Approval] Decision recorded: APPROVED
=====

[Router] Request approved by supervisor → continuing to response.

[Supervisor] Reviewing draft response...
[Supervisor] Draft passes all quality checks ✓
[Supervisor] Notes: Draft approved without modifications.
```

```
[Memory] Saved interaction for customer '3'.
[Supervisor] Interaction saved to memory for customer '3'.

=====

Customer   : Pranay (3)
Query      : I need a refund for my annual subscription.
Intent     : Billing
Department : Billing
Approval   : APPROVED
Supervisor : Draft approved without modifications.
-----
RESPONSE:
-----
Hello Pranay,

Thank you for contacting ABC Technologies Billing Support.

I have received your request regarding: "I need a refund for my annual subscription."

This request requires review and approval from our billing supervisor before it can be processed. Your case has been escalated and you will receive a response within 24 business hours.

Reference information from our policy:

[Source: faq]
Q: How do I request a refund?
A: Submit a refund request through the support portal. Refunds within 30 days are processed automatically; others are reviewed manually within 5 business days.

Best regards,
Jordan | Billing Support Specialist
ABC Technologies Billing

=====

Do you have another query? (yes/no) : no

=====

Thank you for contacting ABC Technologies, Pranay!
Total queries handled this session: 1
Your conversation history is saved under ID: 3

=====

(venv) PS D:\Downloads\customer_support_system\customer_support_system>
```



```

=====
● HUMAN-IN-THE-LOOP APPROVAL REQUIRED
=====
Customer ID : 4
Customer Name: Krishna
Intent      : Billing
Department  : Billing

Customer Query:
  I need a refund for my annual subscription.

Draft Response (for supervisor review):
-----
Hello Krishna,

Thank you for contacting ABC Technologies Billing Support.

I have received your request regarding: "I need a refund for my annual subscription."

This request requires review and approval from our billing supervisor before it can be processed. Your case has been escalated and
you will receive a response within 24 business hours.

Reference information from our policy:

[Source: faq]
Q: How do I request a refund?
A: Submit a refund request through the support portal. Refunds within 30 days are processed automatically; others are reviewed manu-
ally within 5 business days.

Best regards,
Jordan | Billing Support Specialist
ABC Technologies Billing
-----

Supervisor Decision - type 'approve' or 'reject': reject

[Human Approval] Decision recorded: REJECTED
=====

[Router] Request rejected by supervisor -> ending workflow.
[End] Request was rejected by supervisor. Sending rejection response.
[Memory] Saved interaction for customer '4'.

=====

Customer : Krishna (4)
Query    : I need a refund for my annual subscription.

[Human Approval] Decision recorded: REJECTED
=====

[Router] Request rejected by supervisor -> ending workflow.
[End] Request was rejected by supervisor. Sending rejection response.
[Memory] Saved interaction for customer '4'.

=====

Customer : Krishna (4)
Query    : I need a refund for my annual subscription.
Intent   : Billing
Department: Billing
Approval : REJECTED
Supervisor: Request rejected by human supervisor.
-----
RESPONSE:
-----
Hello Krishna,

Thank you for contacting ABC Technologies Support.

After careful review, we are unable to process your request at this time. A support specialist will follow up with you within 24
business hours to discuss your options.

We apologize for any inconvenience and appreciate your patience.

Best regards,
ABC Technologies Support Team
=====

Do you have another query? (yes/no) :

```

QUERY -5:

What was my previous support issue?

```
(venv) PS D:\Downloads\customer_support_system\customer_support_system> python demo.py
[Memory] SQLite database ready at: D:\Downloads\customer_support_system\customer_support_system\memory.db

[System] Starting AI Customer Support System...
[Graph] LangGraph workflow compiled successfully.
[System] System ready.

=====

ABC TECHNOLOGIES – AI CUSTOMER SUPPORT SYSTEM
Built with LangGraph

=====

Please enter your details to begin.

Your name      : Raju
Your customer ID: 2

Welcome, Raju! (ID: 2)
Type your support query below. Type 'quit' to exit.

-----
Your query : What was my previous support issue?

[System] Processing your query...

[Intent] Classifying query: 'What was my previous support issue?'
[Memory] Retrieved 2 history entries for '2'.
[Intent] Classified as: Memory
[Router] Intent='Memory' → routing to 'memory_recall_agent'
[Memory Recall Agent] Retrieving conversation history...

[Supervisor] Reviewing draft response...
[Supervisor] Draft passes all quality checks ✓
[Supervisor] Notes: Draft approved without modifications.
[Memory] Saved interaction for customer '2'.
[Supervisor] Interaction saved to memory for customer '2'.

=====

Customer   : Raju (2)
Query      : What was my previous support issue?
Intent     : Memory
```

Customer : Raju (2)
Query : What was my previous support issue?
Intent : Memory
Department : Memory Recall
Supervisor : Draft approved without modifications.

RESPONSE:

Hello Raju,

I've retrieved your previous support history from our records.

Your most recent support query was:

"My application crashes whenever I upload a file."

Full conversation history:

=== Previous Conversation History ===

[2026-06-27T11:45:33] [Technical Support]

Customer: My application crashes whenever I upload a file.

[2026-06-27T11:45:33] [Technical Support]

Agent: Hello Raju,

Thank you for contacting ABC Technologies Technical Support.

I understand your query: "My application crashes whenever I upload a file."

Based on our knowledge base, here is the information relevant to your request:

[Source: faq]

Q: Why is my application crashing?

A: Common causes include browser cache issues, unsupported file formats, or files exceeding size limits. Try clearing your cache or using a different browser.

--- Additional Troubleshooting ---

[Source: technical_manual]

4. FILE UPLOAD TROUBLESHOOTING

Common causes of file upload crashes:

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- Unsupported file format (supported: PDF, DOCX, XLSX, PNG, JPG, CSV, ZIP)
- Slow internet connection causing timeout
- Browser cache/cookies issue

Please do not hesitate to reach out if you need further assistance.

Best regards,

Sam | Technical Support Engineer
ABC Technologies Technical Support

=== End of History ===

Is there anything else I can help you with today?

Best regards,

ABC Technologies Support Team

=====

Do you have another query? (yes/no) :